Arista Access Network Routing Protocol Flap SOP (KR-ACC-20)

Node: KR-ACC-20 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_5262

Description

A Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on KR-ACC-20, a high-end Arista Access switch, may arise due to suboptimal configuration or transient network conditions. In this SOP, we delve into the intricacies of BGP, OSPF, and ISIS protocol mechanics to identify the root causes of the flap. The BGP route churn can be triggered by frequent IP prefix changes, updates, or withdrawal messages, while OSPF adjacency loss may be caused by issues with OSPF neighbor relationships, router IDs, or interface IP addresses. ISIS LSP flooding, on the other hand, can be precipitated by misconfigurations, topology changes, or hardware issues. Understanding these protocol dynamics is essential to diagnose and resolve the flap effectively.

Pre-requisite Checks

- Verify the site location (KR Puram Access) and ensure that the remote hands are dispatched to the location.

- Check the status of redundant links and ensure that they are functioning properly.

- Confirm that the switch KR-ACC-20 has a stable power supply and temperature.

Potential Root Causes

Cause	Probability
Misconfigured BGP neighbor relationships	High
Transient network conditions (e.g., BGP route churn, OSPF adjacency loss)	Medium
ISIS LSP flooding due to topology changes or hardware issues	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: Displayed BGP summary information, including the number of established BGP connections and the number of prefixes received from each neighbor., Description: This command displays an overview of the BGP connections, which can help identify issues with neighbor relationships or prefix updates.

Command: show ip ospf neighbor, Expected Output: Displayed OSPF neighbor relationships, including the neighbor's IP address, state, and timer information., Description: This command provides a detailed view of the OSPF neighbor relationships, which can help identify issues with OSPF configurations or interface IP addresses.

Command: show isis database, Expected Output: Displayed ISIS LSP database information, including the number of LSPs and the sequence
```

```
numbers., Description: This command displays the ISIS LSP database,
which can help identify issues with LSP flooding or topology changes.
```

Resolution Steps

Step 1: Step: Verify and correct BGP neighbor relationships, Command: show bgp summary | include Established, Description: Verify that the BGP connections are established and correct any misconfigurations.

Step 2: Step: Reset OSPF adjacency, Command: clear ip ospf process, Description: Reset the OSPF process to resolve any issues with OSPF neighbor relationships or interface IP addresses.

Step 3: Step: Clear ISIS LSP database, Command: clear isis database, Description: Clear the ISIS LSP database to resolve any issues with LSP flooding or topology changes.

Post-Check Verification

```
Command: show bgp summary, Expected Output: Established BGP
connections and no prefix updates., Description: Verify that the BGP
connections are established and that there are no prefix updates.

Command: show ip ospf neighbor, Expected Output: Established OSPF
neighbor relationships and no issues with interface IP addresses.,
Description: Verify that the OSPF neighbor relationships are
established and that there are no issues with interface IP addresses.

Command: show isis database, Expected Output: ISIS LSP database is
stable and no issues with LSP flooding., Description: Verify that the
ISIS LSP database is stable and that there are no issues with LSP
flooding.
```

Roll-back Procedures

- Step: Restore the original BGP configuration, Command: copy nvram:startup-config running-config, Description: Restore the original BGP configuration to prevent any configuration drift.

- Step: Reset OSPF adjacency, Command: clear ip ospf process, Description: Reset the OSPF process to resolve any issues with OSPF neighbor relationships or interface IP addresses.

- Step: Rebuild the ISIS LSP database, Command: clear isis database, Description: Rebuild the ISIS LSP database to resolve any issues with LSP flooding or topology changes.